Academic Writing Coach: Week 4 Check-in: Project Plan v1

Track: Writing Assistant
Domain: Academic Writing Support & Feedback
Core HAI Principle: Transparency
Project Type: Interactive AI Writing Mentor
Document Status: Milestone 2 Specification
Date: October 2026

1. Journey Maps

To ensure the Academic Writing Coach meets student needs across different experience tiers, we trace the end-to-end user journey across five distinct stages for our personas:

- **Stage 1 — Start:** The student opens the writing assistant interface with a draft excerpt while experiencing anxiety over upcoming term paper deadlines or complex citation requirements.
- **Stage 2 — Ask:** The student submits a paragraph snippet along with a plain-language query (e.g., "Is my introductory thesis statement clear and properly supported?").
- **Stage 3 — Understand:** The system analyzes the text against `academic_guidelines_dataset.json` and returns constructive feedback paired with explicit style manual references (APA/IEEE) or uncertainty notices.
- **Stage 4 — Act/Fix:** The student reviews the feedback, follows direct style guide links, and revises their draft independently without relying on ungrounded generative rewrites.
- **Stage 5 — Finish:** The student completes their essay revision with total confidence in the formal tone, structural coherence, and citation accuracy of their work.

2. Measurable Objectives (Metrics)

To evaluate system efficacy and strict adherence to our core HAI principle of Transparency, we define five quantifiable target metrics:

- **O1 — Rule & Citation Accuracy:** $\ge 90\%$ of cited style manual rules and reference formats genuinely support the provided feedback when verified against official guidelines.
- **O2 — Task Success:** $\ge 80\%$ student draft improvement rate during usability testing, measured by successful independent revisions.
- **O3 — Transparency Comprehension:** $\ge 90\%$ user comprehension during usability evaluations, verifying that students correctly distinguish mandatory rules from stylistic suggestions.
- **O4 — Performance Latency:** Average end-to-end processing and display latency of less than 2.0 seconds per response.
- **O5 — Uncertainty Handling Rate:** $\ge 95\%$ correct fallback handling on ambiguous stylistic requests or conflicting departmental guidelines, ensuring the system displays an explicit uncertainty notice rather than fabricating rules.

3. HAI Principle Ladder — Transparency

The Transparency ladder establishes four progressive levels of user empowerment in our Human-AI interaction design:

- **Level 1 — Inform:** Upon opening the assistant, the system explicitly greets the user, states that it is an AI writing mentor, and defines its operational scope (assisting with structure and citation rather than drafting essays from scratch). Linked to Task Success (O2).
- **Level 2 — Explain (Rule Attribution):** Every structural critique or citation check includes direct links and references to official style manuals (APA, MLA, IEEE). Linked to Rule Accuracy (O1).
- **Level 3 — Allow Control (Uncertainty Disclosure):** When stylistic rules are open to instructor discretion or departmental guidelines vary, the system displays an explicit uncertainty notice and offers flexible phrasing alternatives rather than rigid mandates. Linked to Uncertainty Handling (O5).
- **Level 4 — Enable Feedback:** Users can submit helpful/unhelpful ratings on whether the writing suggestions were clear and constructive. Linked to Transparency Comprehension (O3).

4. Tool Choices & Technical Architecture

The technical architecture balances high reliability, low latency, and strict grounding:

- **Frontend Interface:** Built with HTML5, CSS3, and JavaScript, featuring accessible card layouts, high contrast for accessibility needs, and screen-reader landmark support.
- **Backend Server:** Powered by Python and FastAPI, handling request routing, payload validation, and rule grounding checks.
- **AI Model & Grounding:** Hosted LLM API (OpenAI/Gemini) configured with strict system prompts enforcing grounded generation against our structured JSON database (`academic_guidelines_dataset.json`).
- **Version Control:** Git repository with structured weekly commits and branch management across all team members.

Technical Architecture Data Flow

Layer	Technology	Function
Frontend	HTML5 / CSS / JS	Responsive chat interface and accessible feedback display
Backend	Python FastAPI	API routing, query parsing, and validation
Grounding	JSON Dataset	Authoritative style rules and citation references
Model	Hosted LLM API	Natural language understanding and constructive critique

5. Evaluation Plan

Our evaluation strategy combines automated verification with empirical user testing:

- **Technical Evaluation:** Automated script testing comparing system feedback against official APA/IEEE style guides to measure rule accuracy and citation precision.
- **Usability Testing:** Peer testing involving at least 5 university students performing structured essay revision tasks.
- **SUS Questionnaire:** System Usability Scale (SUS) surveys administered after testing to evaluate interface intuitiveness and cognitive load.

6. Week 4 Check-in Readiness Checklist

- Detailed journey maps trace user progression across all five core stages.
- Five quantifiable metrics (O1–O5) established with specific percentage targets.
- Transparency ladder outlines four progressive levels linked to core metrics.
- Technical architecture explicitly details frontend, FastAPI backend, and JSON grounding.
- Evaluation plan combines automated rule verification with SUS usability testing.